Stretch Cat

Studio 87. Free and open source (AGPL-3.0-or-later).

Intro

Stretch Cat is a sampler with time stretch. Load a sound; change its speed without changing the pitch, or its pitch without changing the speed, and play it from MIDI keys.

Quick start

- Press Load and pick a sound file (wav, aiff, flac, ogg or mp3).
- Play MIDI notes: the sound is repitched from the Root note. The stretched version is rendered when a control changes.
- Note: this version uses the host's plain parameter view plus a Load button. The designed window (waveform, lanes, slices) is in the mockup in the repository and is not built into the plugin yet.

Stretch

- Stretch mode: Beats, Tones, Texture, Re-Pitch, Complex or Formant+. Beats keeps drums sharp, Tones suits sustained sounds, Texture suits noise and pads, Re-Pitch changes speed and pitch together like a record.
- Speed and pitch linked (Tape): when on, speed and pitch move together.
- Grain, Transients and Smooth shape how the stretch engine cuts and joins the sound.
- Speed: 0.25 to 4 times. Transpose: -24 to +24 semitones. Fine: -100 to +100 cents.

Length and tempo

- Length: Auto (free) or fit the sound to 1/16, 1/8, 1/4, 1/2 note, 1, 2, 4, 8, 16 or 32 bars (4/4). The stretch is set so the sound fills that length at the tempo; the Speed knob is ignored while a length is set.
- Fit tempo (BPM): the tempo used for the length fit, 40 to 240. It follows the host tempo.

Playback

- Playback: Loop, One shot or Gate. Attack up to 500 ms, Release up to 2 s. Root note sets the key where the sound plays unchanged. Gain: -24 to +12 dB. Eight notes can play at once.

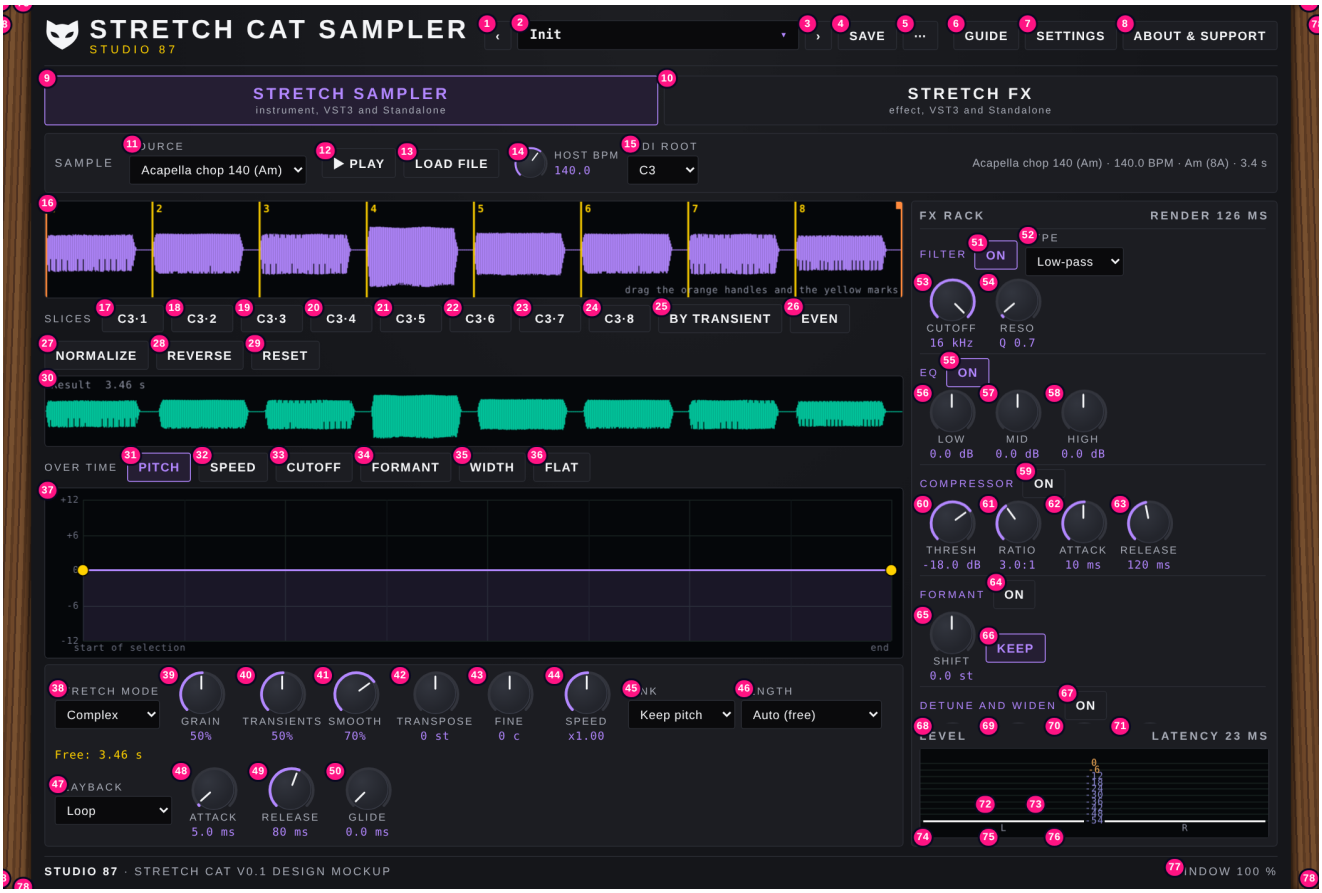
Recommendations

- For loops from a sample pack, set Length to the number of bars and let it fit.
- Use Complex for full mixes, Beats for drum loops and Formant+ for vocals.
- Not tuned by ear yet: please report problems on GitHub.

Interface tour

Every pink number marks a control that has an info bubble in the interactive mockup (docs/mockup), and the tables below carry the text of those bubbles. The pictures come from the mockup, which mirrors the plugin window; the compiled plugin can differ by a few pixels of layout.

STRETCH SAMPLER



Header bar

1	**Previous preset**	Previous preset.
2	**Init**	Open the preset list: factory presets and your own.
3	**Next preset**	Next preset.
4	**Save**	Save the current settings as your own preset (stays on this device).
5	**Delete, export or import your presets**	Delete, export or import your presets.
6	**Guide**	Quick guide: what each part does.
7	**Settings**	Info bubbles and eco mode.
8	**About & Support**	About Studio 87, privacy, and ways to support the project.

Window

9	**Stretch Cat Sampler**	Stretch Cat Sampler: a sampler instrument. Warp a loop, slice it, play slices from your keyboard, transpose over time.
10	**Stretch Cat FX**	Stretch Cat FX: an effect. It grabs the last bars of your DAW audio and lets you stretch it, bend its pitch and perform tape stops, freezes and risers.
11	**Source**	Sampler: the sample that gets warped. FX: the audio your DAW sends in.
12	**Play**	Play the stretched result. Shortcut: Space.
13	**Load file**	Load your own audio file (wav, mp3, flac, ogg). It stays in this browser tab and is never uploaded.

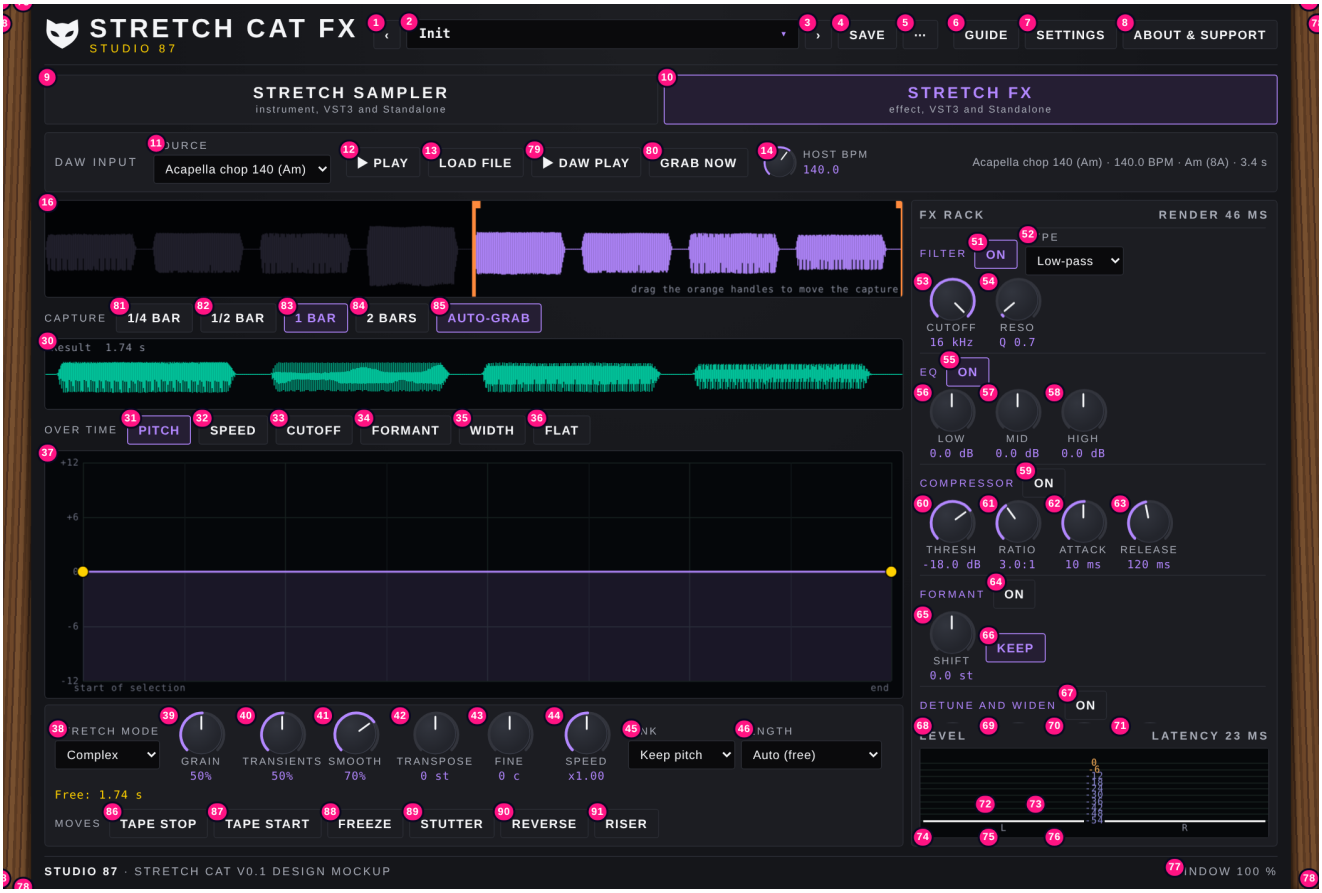
14	**Host BPM**	The tempo your DAW runs at. Keep length stretches the loop to match it. Double-click resets.
15	**MIDI root**	The key that plays the sample at its original pitch.
16	**Source**	The whole sample. The bright part is the selection that gets stretched. Drag the orange handles; in the sampler drag the yellow slice marks.
17	**C3·1**	Slice 1: press to hear it, with your stretch and pitch settings. Keys 1 to 8 do the same.
18	**C3·2**	Slice 2: press to hear it, with your stretch and pitch settings. Keys 1 to 8 do the same.
19	**C3·3**	Slice 3: press to hear it, with your stretch and pitch settings. Keys 1 to 8 do the same.
20	**C3·4**	Slice 4: press to hear it, with your stretch and pitch settings. Keys 1 to 8 do the same.
21	**C3·5**	Slice 5: press to hear it, with your stretch and pitch settings. Keys 1 to 8 do the same.
22	**C3·6**	Slice 6: press to hear it, with your stretch and pitch settings. Keys 1 to 8 do the same.
23	**C3·7**	Slice 7: press to hear it, with your stretch and pitch settings. Keys 1 to 8 do the same.
24	**C3·8**	Slice 8: press to hear it, with your stretch and pitch settings. Keys 1 to 8 do the same.
25	**BY TRANSIENT**	Place slice marks on the strongest hits in the selection.
26	**EVEN**	Cut the selection into eight equal parts.
27	**NORMALIZE**	Make the selection as loud as it can be without clipping (applies to the whole source).
28	**REVERSE**	Play the selection backwards.
29	**RESET**	Reload the original sample.
30	**Result**	What the stretch engine made from the selection, with your pitch, speed and curves. The white line is the play position.
31	**PITCH**	Transposition over the length of the selection. Drag the dots, click to add one, double-click to remove it.
32	**SPEED**	Playback speed over time (with Keep pitch it only changes length, with Tape linked it changes pitch too).
33	**CUTOFF**	Low-pass cutoff over time: close it to make a swell or a wash.
34	**FORMANT**	Formant shift over time. Plugin only: it is drawn here but not audible in this mockup.
35	**WIDTH**	Stereo width over time.
36	**FLAT**	Reset this lane to a straight line.
37	**Curve editor**	Draw what changes over the length of the selection: pitch, speed, cutoff, formant or width. Drag a dot, click empty space to add one, double-click a dot to remove it.
38	**Stretch mode**	In the mockup each mode is a different grain setting. Beats: short grains for drums. Tones: long grains for sustained sounds. Texture: random grains for pads. Re-Pitch: tape speed. Complex and Formant+: all-rounders. The plugin will use dedicated algorithms.
39	**Grain**	Grain size: smaller is tighter in time, larger is smoother. Double-click resets.
40	**Transients**	How hard the engine searches for the best splice point. More keeps hits sharper. Double-click resets.
41	**Smooth**	Length of the crossfade between grains. Double-click resets.
42	**Transpose**	Pitch in semitones. Double-click resets.
43	**Fine**	Pitch in cents. Double-click resets.
44	**Speed**	Overall playback speed. 2 is twice as fast. Double-click resets.
45	**Link**	Keep length: the loop is stretched to your host tempo. Keep pitch: speed changes only the length. Tape linked: speed and pitch move together, like a record.
46	**Length**	Sets how long the stretched sample lasts, in note values or bars at the host tempo (4/4). The stretch fits the selection to that length, so the Speed knob is ignored while a length is chosen. Auto: the length follows Speed and Link.
47	**Playback**	One-shot: plays once. Loop: repeats. Gate: plays while you hold Play. Slices to keys: keys 1 to 8 play the slices.
48	**Attack**	Fade in. Double-click resets.
49	**Release**	Fade out. Double-click resets.
50	**Glide**	Time to slide between notes (plugin only: needs MIDI). Double-click resets.

77	**Window 100 %**	Window size. Drag any edge or corner of the plugin to resize it; the shape stays the same. Double-click here for 100 %.
78	**Resize**	Drag to resize the window. Every edge and corner works, and the shape stays the same.

FX rackrender 126 ms

51	**On**	Turn the filter on or off.
52	**Type**	Filter shape.
53	**Cutoff**	Cutoff frequency. Double-click resets.
54	**Reso**	Resonance. Double-click resets.
55	**On**	Turn the EQ on or off.
56	**Low**	Low shelf (200 Hz). Double-click resets.
57	**Mid**	Mid band (1 kHz). Double-click resets.
58	**High**	High shelf (5 kHz). Double-click resets.
59	**On**	Turn the compressor on or off.
60	**Thresh**	Level where compression starts. Double-click resets.
61	**Ratio**	How hard it squeezes. Double-click resets.
62	**Attack**	How fast it reacts. Double-click resets.
63	**Release**	How fast it lets go. Double-click resets.
64	**On**	Formant control. Plugin only: not audible in this mockup.
65	**Shift**	Natural to chipmunk. Plugin only: not audible in this mockup. Double-click resets.
66	**Keep**	Keep the formants where they were when you transpose. Plugin only.
67	**On**	Turn the detune voices on or off. Takes effect the next time you press Play.
68	**Detune**	Spread of the voices in cents. Double-click resets.
69	**Voices**	How many copies. Double-click resets.
70	**Width**	Haas widening: a delayed copy on the right. Double-click resets.
71	**Haas**	Delay of the widening copy. Double-click resets.
72	**On**	Turn the output stage on or off.
73	**Limit**	Limiter on the output.
74	**Mix**	Blend between the unprocessed stretch (left) and the whole rack (right). Double-click resets.
75	**Gain**	Output gain before the limiter. Double-click resets.
76	**Ceiling**	Limiter ceiling. Double-click resets.

STRETCH FX



Window

79	**DAW play**	Play the unprocessed source like your DAW would. Shortcut: P.
80	**Grab now**	Grab the last bars of the DAW input into the capture buffer. Shortcut: G.
81	**1/4 BAR**	Capture the last 1/4 bar of the DAW input.
82	**1/2 BAR**	Capture the last 1/2 bar of the DAW input.
83	**1 BAR**	Capture the last 1 bar of the DAW input.
84	**2 BARS**	Capture the last 2 bars of the DAW input.
85	**Auto-grab**	Grab a new capture whenever you change the length, and at the start of every bar while the DAW plays.
86	**TAPE STOP**	Hold: the DAW audio slows to a halt like a turntable.
87	**TAPE START**	Hold: the audio spins up from a standstill.
88	**FREEZE**	Hold: freezes a tiny slice of the sound and holds it.
89	**STUTTER**	Hold: repeats the last 1/16 note.
90	**REVERSE**	Hold: plays the DAW audio backwards.
91	**RISER**	Hold: pitch and filter climb for as long as you press.